



The dunes on Mars

The Curiosity rover revealed puzzling aspects of the Namib dunes including ripples not seen on Earth. <https://digit.in/jul20-63>



Into the Lost City

A deep dive into the Lost City Hydrothermal Field, and investigating the microbial life found there. <https://digit.in/jul20-64>

shaped structure called the Atlantis Massif. Towards the south eastern part of this dome is a field of hydrothermal vents known as the Lost City. Around the vents are towers of carbonate, which look like spires of some alien metropolis, which makes the name appropriate. The hydrothermal field was discovered only recently, in 2000. These deep sea hydrothermal vents represent the last major ecosystem discovered on the planet.



A chimney at the Lost City

The creatures living around these vents are really strange and include eyeless shrimps with specialised organs on their backs that can distinguish between dark and light surroundings, or worms without a digestive tracts that get their nutrition from colonies of bacteria. The vents are so deep that the sunlight does not reach them, so photosynthesis is not a viable approach for energy production. Instead, the creatures living here depend on chemosynthesis, subsisting on the chemicals expelled by the vents, mainly hydrogen gas. These alkaline hydrothermal vents are where scientists believe that the first life forms on the planet sprung up. There are two other competing theories, where life emerged from a primordial soup because of chemical reactions, or life was carried to the planet through interplanetary or even interstellar objects from another location, a process called panspermia. The conditions on ancient seas were very different from today, and scientists recreated these conditions to check if life could emerge around the hydrothermal vents. They

found that the chemical compounds necessary for life to spring up, could indeed be formed around these ancient hydrothermal vents. Vents very similar to the Lost City may be present on the moons of Enceladus and Saturn, on the sea floor of the subsurface ocean, below the ice cap on top. If this is the case, then both these moons would have all the ingredients to support life, as well as the right conditions. Accessing these vents would be a difficult prospect though, as it would mean drilling beneath the ice shell, and then using an underwater rover. However, future missions flying through the plumes of water erupting from these moons can perhaps study the chemical composition of the subsurface ocean.

PENITENTES

Penitentes are snow formations in high altitudes that are blades of ice oriented towards the Sun. As the sunlight usually falls in an angle, these blades of snow are angled slightly forward, and appear like people praying or performing penance, which is how the formations get their name. These penitentes can cover entire fields and were first reported in scientific literature by Charles Darwin, high in the Andes mountains. The height of the features vary vastly on Earth, and they can range from a few centimeters high to a towering five meters. These penitentes can be found in the high altitude Atacama desert, across Chile and Argentina. It can be a strange sight, to see blades of white snow sticking out of the brown sands of the desert. The tallest penitentes are formed in the dry deserts of the tropics, because of the weather conditions. While wind does play a role in eroding these structures, they are primarily shaped by the action of the sunlight. The formations are made of a combination of ice and snow, with finer grains towards the top. The formations occur where the

dew point, or the temperature at which the air can no longer hold any more water vapour, is below the freezing point of water. The conditions cause the water to sublime, or turn into vapour from solid without melting first. The accumulations of these features are referred to as penitentes gardens.

For penitentes to form, you need low temperatures, plenty of exposure to sunlight, and dry air. All these conditions are present in the equatorial region of Europa, the Jovian moon. Scientists used the data from the Galileo spacecraft to come to the conclusion that Europa too, had penitentes just like the Earth. If they are present, the penitentes on Europa are expected to be much larger than the similar features on Earth, as tall as five storey buildings. Each of the blades could be fifteen meters tall and 7 meters wide. Apart from Europa, these penitentes are also found on Pluto, in what is known as the bladed terrain. The New Horizons spacecraft made the discovery. On Pluto as well, the bladed terrain is found in the equatorial regions. Unlike Earth though, the penitentes



Penitentes in Chile

on Pluto are made up of methane ice, instead of water ice. While the formations on Pluto have been directly imaged by New Horizons, the formations on Europa are only speculated. Future missions will have to confirm the presence of these penitentes on Europa. Any spacecraft attempting to land in the equatorial regions, will have to be careful to avoid these icy spikes, or will get shredded as it approaches the surface. 